Bayes Filter and Kalman Filter for State Estimation in a Driverless Vehicle

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Abstract*— Accurate state estimation is essential for the safe and reliable operation of an autonomous vehicle. A vehicle cannot directly determine its true position, velocity, and orientation at every instant because its sensors are affected by measurement noise, while its motion model is subject to uncertainties. Bayesian filtering provides a mathematical framework for estimating the vehicle's state by combining predictions from a motion model with information obtained from sensor measurements. The Kalman Filter is a special case of the Bayes filter that provides an efficient solution when the system is linear, and its uncertainties follow Gaussian distributions.

Keywords*— Autonomous vehicles, Bayesian filtering, Kalman Filter, state estimation, sensor fusion, Gaussian distribution, Extended Kalman Filter.

I. Introduction

A driverless vehicle must continuously estimate its position, velocity, orientation, and other parameters to navigate safely and accurately. However, the true state of a vehicle cannot be measured directly. Instead, the vehicle relies on sensors such as Global Navigation Satellite Systems (GNSS), Inertial Measurement Units (IMUs), wheel-speed sensors, cameras, and LiDAR. Each sensor provides measurements that contain some degree of uncertainty or noise.

The vehicle also uses a mathematical motion model to predict how its state changes over time. However, this model cannot perfectly represent real-world behaviour because of factors such as wheel slip, changing road conditions, and inaccuracies in control inputs. Therefore, relying only on predicted motion or individual sensor readings can result in an inaccurate estimate. Filtering algorithms address this problem by combining the predicted state with sensor measurements to obtain a more reliable estimate.

Bayesian filtering provides a general mathematical framework for this process. The Kalman Filter is one of its most widely used implementations, particularly for systems that can be represented using linear models and Gaussian noise. These methods form an important foundation for localisation and sensor fusion in autonomous vehicles.

II. Bayes Filter

A. Fundamental Concept

The Bayes filter is a probabilistic method used to estimate the state of a system over time. Instead of assuming that the system has one perfectly known state, it maintains a *belief*, which is a probability distribution representing how likely different states are. Let x_t represent the state of the vehicle at time t, u_t the control input, and z_t the sensor measurement. The objective is to calculate the probability of the current state given all available measurements and control inputs. This is represented as:

$$bel(x_t) = p(\, x_t \mid z_{1:t}, u_{1:t}\,)$$

Here, bel(x_t) is the current belief about the vehicle's state, while $z_{1:t}$ and $u_{1:t}$ represent all measurements and control inputs up to time t.

The Bayes filter updates this belief through two main stages: prediction and measurement update.

1) Prediction

The prediction step estimates the vehicle's next state using its previous state and the motion model. The motion model describes how the vehicle is expected to move when a particular control input is applied. Since the actual movement may differ from the prediction, the model also accounts for uncertainty. The predicted belief is calculated using:

$$\overline{bel(x_t)} = \int^{p} (\, x_t \mid x_{t-1}, u_t\,) bel\big(x_{\{t-1\}}\big) dx_{t-1}$$

In this equation, $\overline{bel(x_t)}$ represents the predicted belief before considering the latest sensor measurement. The term $p(\, x_t \mid x_{t-1}, u_t\,)$ is the motion model, which describes the probability of reaching the current state from the previous state under the applied control input. The integral combines the possible previous states and their probabilities to obtain the predicted distribution.

2) Measurement Update

After predicting the next state, the filter incorporates the latest sensor measurement to correct its estimate. The measurement likelihood describes how probable the observed sensor reading is for a particular estimated state. The updated belief is obtained using Bayes' theorem:

$$bel(x_t) = \eta p(z_t|x_t)\,\overline{bel(x_t)}$$

Here, $p(z_t|x_t)$ is the measurement likelihood, which indicates how likely the observed measurement z_t is if the vehicle is in state x_t. The term η is the normalising constant, which ensures that the resulting probability distribution integrates to one. The updated belief $bel(x_t)$ is therefore the posterior estimate after incorporating the sensor information. The normalising constant can be expressed as:

$$\eta = \frac{1}{\int^{p}(z_t \mid x_t)\overline{bel(x_t)}\,dx_t}$$

For example, if the motion model predicts that the vehicle is at a particular position but GNSS reports a slightly different position, the measurement update combines both sources of information. The final estimate depends on the uncertainty associated with the prediction and the sensor measurement.

Thus, the Bayes filter continuously alternates between predicting the state using the motion model and correcting that prediction using sensor measurements.

III. Assumptions for the Kalman Filter

The general Bayes filter can represent complex probability distributions, but calculating and maintaining these distributions can be computationally expensive. The Kalman Filter simplifies this process by introducing specific assumptions about the system and its uncertainties. Under these assumptions, the belief distribution remains Gaussian and can be represented using only its mean and covariance.

A. Linear System Models

The Kalman Filter assumes that the system's state transition and measurement relationships are linear. The state transition model is represented as:

$$x_t = Fx_{t-1} + Bu_t + w_t$$

Here, x_t is the current state vector and x_{t-1} is the previous state vector. F is the state transition matrix, which describes how the previous state influences the current state. B is the control input matrix, u_t represents the applied control input, and w_t represents process noise or uncertainty in the motion model. The measurement model is expressed as:

$$z_t = Hx_t + v_t$$

In this equation, z_t is the sensor measurement, H is the observation matrix that maps the system state to the measurement space, and v_t represents measurement noise.

B. Gaussian Noise and Initial Belief

The Kalman Filter assumes that process noise, measurement noise, and the initial state uncertainty follow Gaussian distributions. A Gaussian distribution is a continuous probability distribution characterised by a symmetric, bell-shaped curve. It is described by its mean, which determines the centre of the distribution, and its variance, which determines its spread. A smaller variance produces a narrower curve, indicating greater certainty, while a larger variance produces a wider curve, indicating greater uncertainty. These assumptions are represented as:

$$w_t \sim N(0,\, Q)$$

$$v_t \sim N(0,\, R)$$
$$x_0 \sim N(\widehat{x_0},\, P_0)$$

Here, N denotes a Gaussian distribution, with zero mean for the noise terms. Q is the process noise covariance matrix, which describes uncertainty in the motion model, while R is the measurement noise covariance matrix, which describes sensor uncertainty. The initial state is represented by its estimated mean ($\hat{x}_0$) and covariance (P_0).

C. Why these assumptions matter

Linear transformations of Gaussian distributions remain Gaussian. Consequently, when the system has linear state and measurement models and Gaussian noise, the prediction and measurement update steps preserve the Gaussian form of the belief distribution. The filter therefore only needs to calculate the estimated state mean and its covariance rather than maintaining an entire probability distribution.

These assumptions make the mathematical calculations efficient and suitable for real-time applications. However, actual vehicle dynamics are often nonlinear. For instance, the vehicle's position changes according to its velocity and heading through trigonometric relationships. Sensor errors may also be non-Gaussian or contain outliers. When these assumptions are not satisfied, a standard Kalman Filter may produce inaccurate estimates. Variants such as the Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), and Particle Filter are used to handle more complex systems.

IV. Kalman Filter

The Kalman Filter is a recursive state estimation algorithm that combines predictions from a mathematical system model with measurements obtained from sensors. It maintains two important quantities: the estimated state vector $\hat{x}$, representing the estimated state of the system, and the covariance matrix P, representing the uncertainty associated with that estimate. Its operation can be divided into five steps.

A. State Prediction

The first step predicts the next state using the previous corrected estimate and the control input. The state prediction equation is:

$$\hat{x}_{t|t-1} = F\,\hat{x}_{t-1|t-1} + B\,u_t$$

Here, $\hat{x}_{t|t-1}$ is the predicted state at time t, before incorporating the latest sensor measurement. F is the state transition matrix, B is the control input matrix, and u_t represents the applied control input. This equation uses the motion model to estimate how the system is expected to move during the next time interval.

B. Covariance Prediction

Along with predicting the state, the filter also predicts how uncertain that estimate is. This is calculated using:

$$P_{t|t-1} = F\,P_{t-1|t-1}\,F^T + Q$$

Here, $P_{t|t-1}$ is the predicted covariance matrix, $P_{t-1|t-1}$ is the previous corrected covariance, and F^T is the transpose of the state transition matrix. Q represents the process noise covariance, accounting for uncertainty in the motion model. Since errors can accumulate during prediction, the uncertainty generally increases as the system evolves.

C. Kalman Gain Calculation

The Kalman gain determines how much importance should be assigned to the predicted state compared with the new sensor measurement. It is calculated as:

$$K_t = P_{t|t-1}\,H^T\,S_t^{-1}$$

Here, K_t is the Kalman gain, H is the observation matrix, $P_{t|t-1}$ is the predicted covariance, and R is the measurement noise covariance. The gain depends on the relative uncertainty of the prediction and the sensor measurement.

The innovation covariance, represented by S_t, describes the uncertainty associated with the difference between the actual sensor measurement and the predicted measurement. It combines the uncertainty in the predicted state, transformed into measurement space, with the sensor measurement noise.

$$S_t = H\,P_{t|t-1}\,H^T + R$$

If a sensor is very noisy, its measurement uncertainty R is high, so the filter generally gives less importance to that measurement and relies more on the predicted state. Conversely, if a sensor is highly accurate, its measurement uncertainty is low, allowing the filter to give greater importance to the new measurement. The Kalman gain is recalculated at every time step rather than remaining constant.

D. State Update

After calculating the Kalman gain, the filter compares the actual sensor measurement with the measurement predicted from the estimated state. Their difference is called the innovation or measurement residual:

$$y_t = z_t - H\,\hat{x}_{t|t-1}$$

Here, z_t is the actual sensor measurement, while $H\hat{x}_{t|t-1}$ is the predicted measurement. The innovation represents the discrepancy between the predicted and observed values. The predicted state is then corrected using:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t\,y_t$$

The filter adds a weighted portion of the innovation to the predicted state. The Kalman gain controls the magnitude of this correction, balancing the information obtained from the motion model and the sensor measurement. The result is the updated state estimate.

E. Covariance Update

The final step updates the covariance matrix to reflect the uncertainty remaining after incorporating the sensor measurement. The Joseph form of this update is:

$$P_{t|t} = (I-K_t\,H)\,P_{t|t-1}\,(I-K_t\,H)^T + K_t R K_t^T$$

Here, $P_{t\,|\,t}$ is the updated covariance matrix and I is the identity matrix. This equation accounts for the uncertainty remaining in the corrected estimate and helps preserve the numerical properties of the covariance matrix. The updated state estimate and covariance are then used as inputs for the next prediction cycle, making the Kalman Filter a continuous and recursive process.

V. Relevance to a Driverless Vehicle

A. Role in the Driverless Pipeline

In a driverless vehicle, a Kalman Filter is primarily used in the state estimation and sensor fusion stage. The general processing pipeline consists of sensor data acquisition, state estimation, perception, path planning, and vehicle control. State estimation provides the vehicle with an updated estimate of its position, velocity, and orientation, which is essential for making reliable driving decisions.

Different sensors provide complementary information. GNSS provides global position and velocity estimates but may be affected by signal blockage and multipath errors. An IMU measures linear acceleration and angular velocity, allowing rapid motion tracking, but small measurement biases can accumulate into significant drift over time. Wheel-speed sensors provide information about vehicle motion, although wheel slip can affect their accuracy.

Cameras and LiDAR provide information about the surrounding environment, such as road boundaries, obstacles, and track landmarks. Their processed outputs can support perception and localisation. When appropriate measurements are available, these observations can also contribute to corrections in a state estimation system. Combining complementary sensor information produces a more reliable estimate than depending on a single sensor.

B. Applications of the Extended Kalman Filter

The motion of a real vehicle is nonlinear because its position depends on its velocity and heading. For example, a simplified vehicle motion model can be expressed as:

$$x_t = x_{t-1} + v_{t-1}\cos(\theta_{t-1})\,\Delta t$$

$$y_t = y_{t-1} + v_{t-1}\sin(\theta_{t-1})\,\Delta t$$

$$\theta_t = \theta_{t-1} + \omega_{t-1}\Delta t$$

Here, x_t and y_t represent the vehicle's position, v_t is its velocity, θ_t is its heading angle, ω_t is its yaw rate, and Δt is the time interval. The sine and cosine terms make the motion model nonlinear, meaning that the standard linear Kalman Filter cannot directly represent the complete model.

The Extended Kalman Filter (EKF) addresses this limitation by locally linearising nonlinear system and measurement models using Jacobian matrices that provide local linear approximations around the current estimated state. It then applies the Kalman filtering prediction and correction steps using these local linear approximations.

By combining motion predictions with suitable sensor measurements, a Kalman Filter or EKF can reduce estimation errors and provide a more stable estimate of the vehicle's state. This estimate can then be used by the planning and control modules to determine the vehicle's trajectory and execute steering, acceleration, and braking commands.

VI. Conclusion

Bayesian filtering provides a general framework for estimating the state of a system by combining predictions from a motion model with sensor measurements. The Kalman Filter simplifies this process under linearity and Gaussian uncertainty assumptions, maintaining the estimated state and its covariance through recursive prediction and correction steps. The Kalman gain determines the relative influence of the predicted state and the latest measurement based on their uncertainties.

In driverless vehicles, accurate state estimation is essential for localisation, sensor fusion, planning, and control. Since actual vehicle motion is nonlinear, the Extended Kalman Filter is often more suitable than the standard linear Kalman Filter. By combining information from sensors such as GNSS, IMUs, and wheel-speed sensors, filtering algorithms help provide the reliable state estimates required for autonomous navigation.

References

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